LIN7076 – Foundations of Computational Linguistics

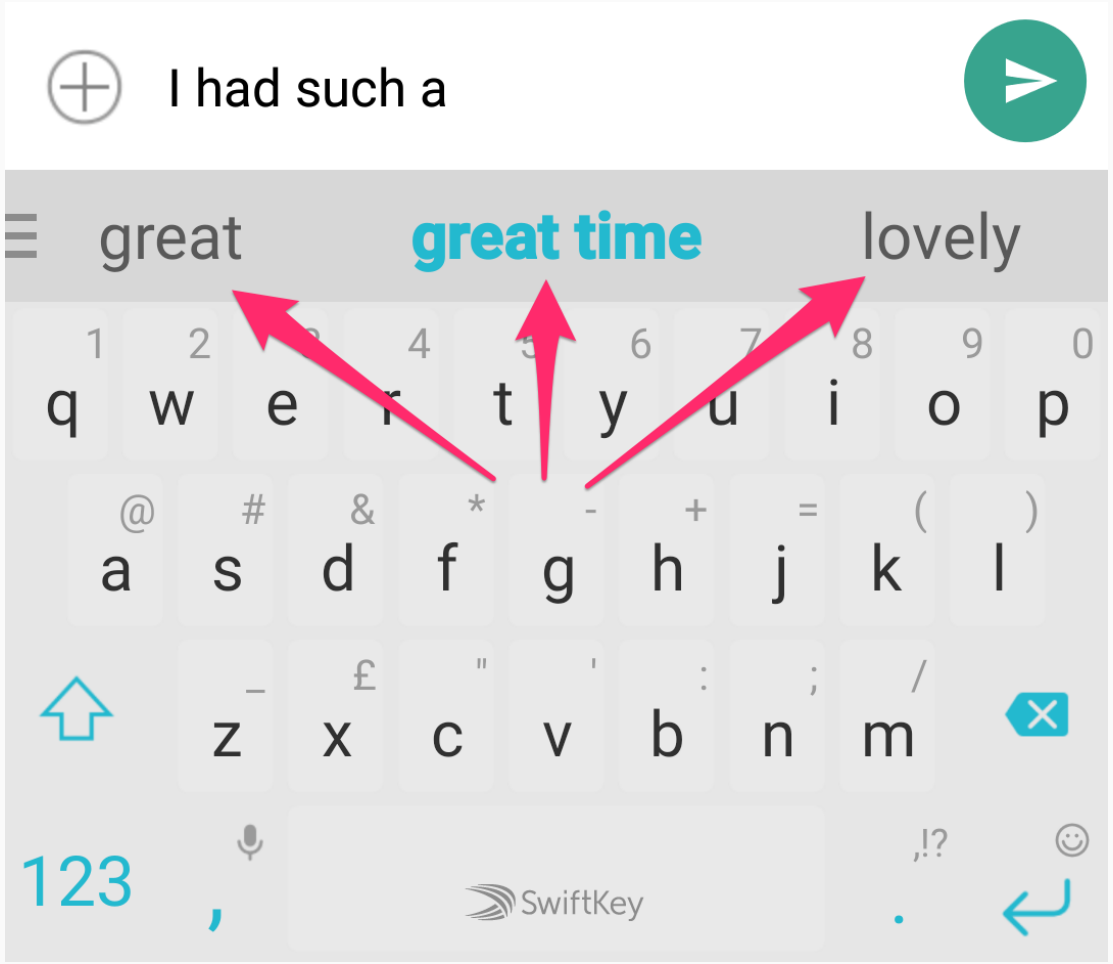
N-gram models

Adèle Hénot-Mortier

28/09/2026

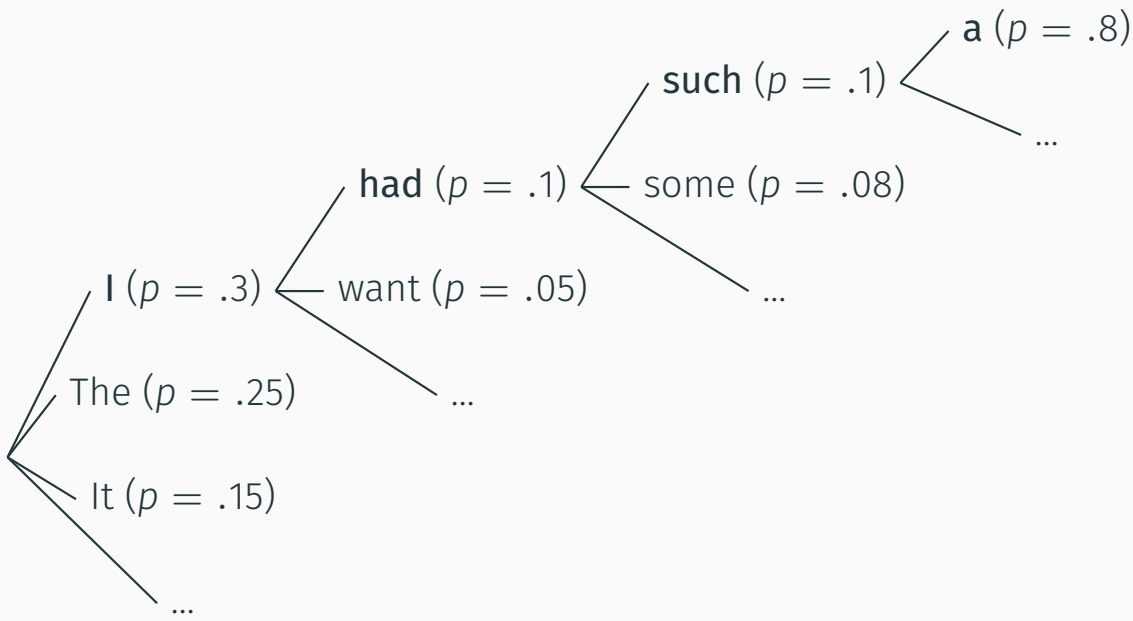
Queen Mary University of London

Generating text



The intuition behind n-grams

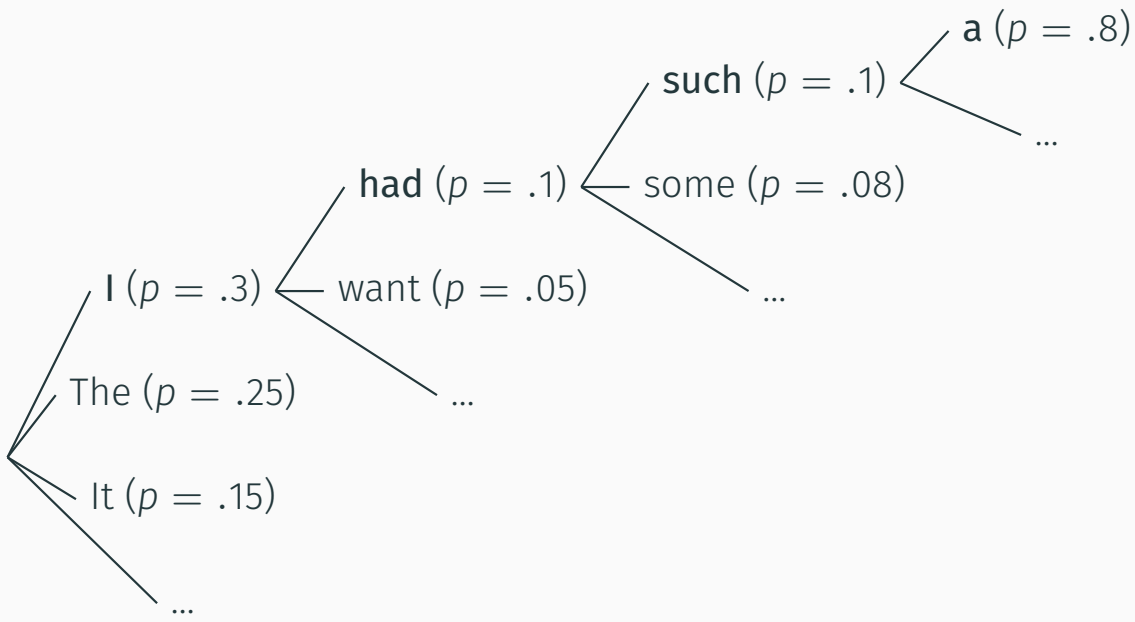
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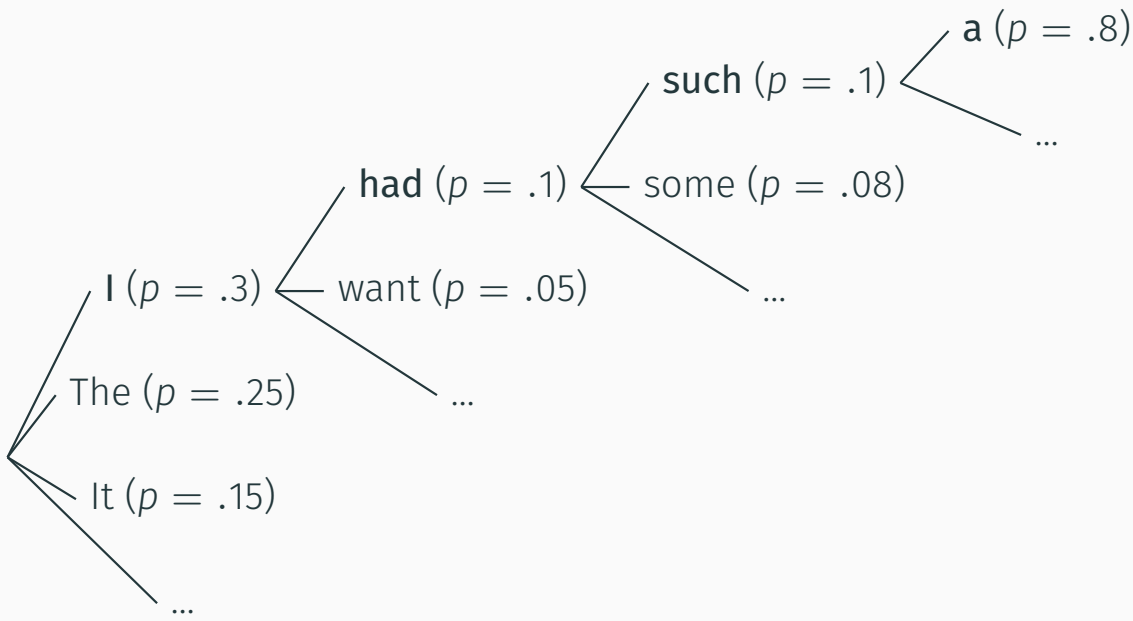
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Plan for today

- Understand the *n*-gram **prediction process**.
- Understand how the probabilities used in that process are **approximated** from a training corpus.
- Understand how the resulting model can be **evaluated** against a test corpus.
- Explore **fixes/improvements** of shortcomings of the basic model.

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The n-gram generation process

Why n-gram?

When predicting a word, pick the **most likely, given wat was written before** within a specific window.

- We want to compute the conditional probability of a word w_i, given a bunch of preceding words $[w_{i-n+1}, ..., w_{i-1}]$, that we call the **context**, of size $n-1$.
- The sequence $[w_{i-n+1}, ..., w_{i-1}] + [w_i]$ is of size n. We call it a **n-gram**.
  - For instance, the sequence $[I, had, such, a]$, is a 4-gram.
- A **n-gram model**, predicts the next word, given a $(n-1)$-gram of preceding words.
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Refresher on conditional probabilities

- The **conditional probability** of an event A, given an event B, is the probability of A when B is known to happen.
- It can be computed using the joint probability of A and B, and B's probability.

conditional probability of B given A

$$\overbrace{P(A|B)} = \frac{\overbrace{P(A,B)}^{\text{joint probability of A and B}}}{P(B)} = \frac{\overbrace{P(A \cap B)}^{\text{probability A and B occcur together}}}{P(B)}$$

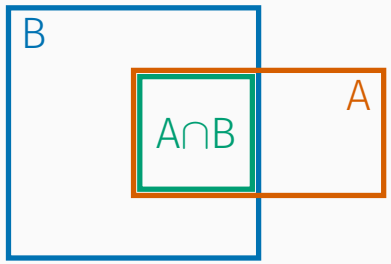
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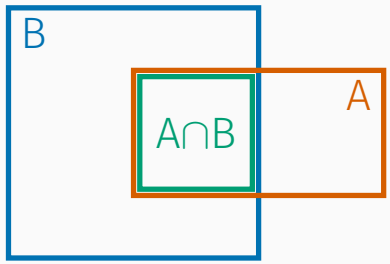
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joint probability of A and B

probability A and B occcur together



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Quick example

- Suppose our sentence is a polar (=yes-no) question and our first word is either *Are* (with probability 25%) or *Is* (with probability 75%).
- Suppose moreover that [*Are the*] is a sequence with 10% probability and [*Is the*] a sequence with 15% probability.

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- What's the conditional probability of *the* given *Are*? Here, A is the event that the second word is *the*, while B is the event that the first word is *Are*, and we compute
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Rephrasing the n-gram intuition using conditional probabilities

- Our target even A, is that of the word w_i being used. We call this word target.
- Our given event B, is that of the words $[w_{i-n+1}, ..., w_{i-1}]$ being used before. We call this sequence context.
- We want to compute $P(A|B)$, the probability of wi given $[w_{i-n+1}, ..., w_{i-1}]$!

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- $P(w_i|[w_{i-n+1}, ..., w_{i-1}])$ depends solely on the context, and the context appended with the target (= the n-gram!).

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- What would a **unigram model** look like? In a unigram model, we predict the next word based on a context of size 0 (=1 − 1). So our conditional probability is $P(w_i|[]) = P(w_i)$ (sometimes called "prior"). It is fully oblivious of the left context.

A 3-gram example with made-up probabilities

- Suppose our sentence starts with the bigram *I had*. Suppose that *such* and *some* are the only 2 possible continuations.

$$P(\textit{such}|\textit{I had}) = \frac{P(\textit{I had such})}{P(\textit{I had})} = .7$$

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(3) The *student* who Jo said that Ed thinks that ... Al likes __ ...

student: *subject*

that Ed thinks that ... Al likes __ ...: arbitrarily long, nested relative clauses

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- Could a *n*-gram model with a fixed window size correctly predict the number marking on the following verb? Could it even predict that it should be a verb?

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Why windows, anyway

- Using a fixed window makes the assumption that most of the time, language generation is oblivious of what was said too long ago. As we've just seen, this is not always true.
- This is known as the **Markov assumption**, which applies to processes well beyond CompLing! Biology, economy, robotics...
- Even if imperfect, the Markov assumption is computationally more efficient, as it require us to compute n-gram and $n-1$-gram **occurrences** for a fixed n, as we will soon see.
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A note on sampling

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- What if two possible targets w_i and w_i' are associated to very close conditional probabilities, e.g. .999 and .1? Should we still pick the most likely, all the time?

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- In practice, we **sample**: we pick target words randomly, giving more chances to the more likely ones.
- One simple way to think about it: drawing one ball from a big lottery box such that each target word is assigned a ball color, and gets as many balls as its conditional probability converted to a round percentage (so $p = .155$ means 16 balls).



Sampling example

- Remember our made up conditional probabilities for *such* and *some*, given the context *I had*?

$$P(\textit{such}|\textit{I had}) = \frac{P(\textit{I had such})}{P(\textit{I had})} = .7$$

$$P(\textit{some}|\textit{I had}) = \frac{P(\textit{I had some})}{P(\textit{I had})} = .3$$

- Instead of returning *such* (presumed most likely) 100% of the time, sampling will return *such* 70% of the time, and *some* 30% of the time.
- More generally, any target word w would be picked $(100 \times P(w|\textit{I had}))\%$ of the time.

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Approximating conditional probabilities

From probabilities to counts

- So far we have reasoned about hypothetical probabilities of words or sequences of words. **How are these probabilities estimated?**
- The probability of a sequence of words S of any fixed length (can be 1, i.e. S is a single word) can be estimated based on a training corpus.
- This is done by counting the number of S's occurrences, and dividing it up by the sum of the counts of all same-length sequences.
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From probabilities to counts

- [The following demontration is a bit advanced; you will not be asked to reproduce or use it in assessments]. Let $C(S)$ be the number of occurrences of S in a given corpus.

$$P(S) = \frac{C(S)}{\sum_{|S'|=|S|} C(S')} \triangleq \alpha C(S)$$

- Suppose now S' is obtained from S by removing S's last word: $S = S' + w$. If S is a n-gram, S' is a $(n-1)$-gram. S''s probability can be expressed by summing the probabilities of its immediate continuations...

$$P(S') = \sum_w P(S' + w) = \sum_w \alpha C(S' + w) = \alpha \sum_w C(S' + w) = \alpha C(S')$$

- We have just expressed probabilities of n and $(n-1)$-grams using their **counts** and the **same normalizing factor** α!

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From conditional probabilities to counts

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P([w_{i-n+1}, \ldots, w_i]) &= \alpha C([w_{i-n+1}, \ldots, w_i]) \\
P([w_{i-n+1}, \ldots, w_{i-1}]) &= \alpha C([w_{i-n+1}, \ldots, w_{i-1}])
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- Plugging this into our n-gram prediction formula, the αs eventually cancel out! So we get a conditional probability that **solely depends on n-gram counts**.

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P(w_i | [w_{i-n+1}, \ldots, w_{i-1}]) &= \frac{P([w_{i-n+1}, \ldots, w_i])}{P([w_{i-n+1}, \ldots, w_{i-1}])} \\
&\simeq \frac{C([w_{i-n+1}, \ldots, w_i])}{C([w_{i-n+1}, \ldots, w_{i-1}])} \\
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Quick example

- Suppose our corpus only contains polar questions whose first word is either *Are* (1050 occurrences) or *Is* (3150 occurrences).
- Suppose moreover that [*Are the*] is a sequence occurring 420 times sentence-initially in the corpus and [*Is the*] occurs 630 times, also sentence-initially.

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- These conditional probabilities should look familiar: it's because I made the count s.t. they correspond to the probabilities of the previous exercise $\times 4200$ (i.e. $\alpha = {}^{1}/_{4200}$).

Dealing with sentence boundaries

$$P(w_i|[w_{i-n+1}, ..., w_{i-1}]) \simeq \frac{C([w_{i-n+1}, ..., w_i])}{C([w_{i-n+1}, ..., w_{i-1}])} = \frac{C(context + target)}{C(context)}$$

- The above equation predicts the next word, given a context of size $n-1$. But what should we do at the very beginning of a sentence, when the context is empty? And when should we stop?
- To properly initiate and terminate generation, we posit two extra silent "words": **<s>** ("**beginning** of sentence") and **</s>** ("**end** of sentence").
- For a n-gram model, each sentence in the training corpus gets **padded with $n-1$ <s> at the beginning, and 1 </s> at the end**. Counts incorporate these extra words.
- To start generating a sentence, we can then assume an initial, silent context made of $n-1$ <s> symbols!
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- And we stop as soon as **</s>** gets generated.

Estimating conditional probabilities: a worked example

- Let's compute a toy bigram model. We pad with 1 <s> symbol.

(5) a. <s> my dog is nice </s>

b. <s> my cat is nasty </s>

c. <s> your cat hates my cat </s>

Estimating conditional probabilities: a worked example

- Let's compute a toy bigram model. We pad with 1 <s> symbol.

(6) a. <s> my dog is nice </s>

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- Bigrams: (<s>, my), (my, dog), (dog, is), (is, nice), (nice, </s>), (<s>, my), (my, cat), (cat, is), (is, nasty), (nasty, </s>), (<s>, Your), (Your, cat), (cat, hates), (hates, my), (my, cat), (cat, </s>).
- Unigrams: <s>, my, dog, is, nice, </s>, <s>, my, cat, is, nasty, </s>, <s>, Your, cat, hates, my, cat, </s>.

Estimating conditional probabilities: a worked example

(7) a. <s> my dog is nice </s>

b. <s> my cat is nasty </s>

c. <s> your cat hates my cat </s>

$w_i \rightarrow$ $\downarrow w_{i-1}$	my	dog	is	nice	cat	nasty	your	hates	</s>	Total
<s>	2	0	0	0	0	0	1	0	0	3
my	0	1	0	0	2	0	0	0	0	3
dog	0	0	1	0	0	0	0	0	0	1
is	0	0	0	1	0	1	0	0	0	2
nice	0	0	0	0	0	0	0	0	1	1
cat	0	0	1	0	0	0	0	1	1	3
nasty	0	0	0	0	0	0	0	0	1	1
your	0	0	0	0	1	0	0	0	0	1
hates	1	0	0	0	0	0	0	0	0	1
Total	3	1	2	1	3	1	1	1	3	

$$P(my|\texttt{<s>}) = \frac{C(\texttt{<s>}my)}{C(\texttt{<s>})} = \frac{2}{3}$$

$w_i \rightarrow$ $\downarrow w_{i-1}$	my	dog	is	nice	cat	nasty	your	hates	</s>	Total
<s>	2	0	0	0	0	0	1	0	0	3
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your	0	0	0	0	1	0	0	0	0	1
hates	1	0	0	0	0	0	0	0	0	1
Total	3	1	2	1	3	1	1	1	3	

$$P(dog|my) = \frac{C(\text{my dog})}{C(my)} = \frac{1}{3}$$

$w_i \rightarrow$ $\downarrow w_{i-1}$	my	dog	is	nice	cat	nasty	your	hates	</s>	Total
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is	0	0	0	1	0	1	0	0	0	2
nice	0	0	0	0	0	0	0	0	1	1
cat	0	0	1	0	0	0	0	1	1	3
nasty	0	0	0	0	0	0	0	0	1	1
your	0	0	0	0	1	0	0	0	0	1
hates	1	0	0	0	0	0	0	0	0	1
Total	3	1	2	1	3	1	1	1	3	

$$P(hates|cat) = \frac{C(\text{cat hates})}{C(cat)} = \frac{1}{3}$$

$w_i \rightarrow$ $\downarrow w_{i-1}$	my	dog	is	nice	cat	nasty	your	hates	</s>	Total
<s>	2	0	0	0	0	0	1	0	0	3
my	0	1	0	0	2	0	0	0	0	3
dog	0	0	1	0	0	0	0	0	0	1
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your	0	0	0	0	1	0	0	0	0	1
hates	1	0	0	0	0	0	0	0	0	-
Total	3	1	2	1	3	1	1	1	3	

- Can you guess which sentence(s) would be most likely to be generated by this bigram model?

Bonus: Evaluating n-gram models

Probability of a corpus

- To evaluate how good a n-gram model is, one needs new data: typically a test corpus that is not too different from the training corpus.[2]
- We can then evaluate how likely this whole corpus is to be generated by our n-gram model!
- This is thanks to the chain rule:

$$P(A_1 \cap A_2 \cap \ldots \cap A_k) = P(A_1) \times \prod_{i=2}^{k} P(A_i | A_1 \cap \ldots \cap A_{i-1})$$

$$\overset{Markov}{=} P(A_1) \times \prod_{i=2}^{k} P(A_i | A_{i-n+1} \cap \ldots \cap A_{i-1})$$

$$P([w_1 \ldots w_k]) \overset{Markov}{=} P(w_1) \times \prod_{i=2}^{k} P(w_i | [w_{n-i+1} \ldots w_{i-1}])$$

[2] Why not test on the training corpus?

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[2] Why not test on the training corpus?

From probability to perplexity

- Measuring the performance of a n-gram based on the probability of a test corpus is nice, but depends on the corpus size: bigger corpora will mechanically be less probable (why?).
- To avoid this issue, we use a normalized variant of corpus probability called **perplexity**. The lower the perplexity, the better the model.

$$PPL([w_1 ... w_k]) = P([w_1 ... w_k])^{-\frac{1}{k}}$$

$$\overset{Markov}{=} \left(P(w_1) \times \prod_{i=2}^{k} P(w_i | [w_{n-i+1} ... w_{i-1}]) \right)^{-\frac{1}{k}}$$

- If $k \uparrow$, $\frac{1}{k} \downarrow$, $-\frac{1}{k} \uparrow$ and $p^{-\frac{1}{k}} \downarrow$ (with $p \geq 0$). So if two corpora have same probability, the larger one will get a smaller perplexity.

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An issue with perplexity

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- Looking at the above product of probabilities, we notice that if only one probability is 0, the whole product is $\frac{1}{0} = +\infty$!
- This would happen if the test corpus contains **one *n*-gram that the training corpus does not have** – which is far from unlikely!
- So perplexity is **unlikely to distinguish good from bad *n*-gram models**, as we defined them. Both kinds will likely get an infinite perplexity.
- More generally, probabilities learned from a finite training corpus will necessarily be 0 for many perfectly reasonable *n*-grams...bringing us to the **competence-performance distinction**.

An issue with perplexity

$$PPL([w_1 ... w_k]) \overset{Markov}{=} \left(P(w_1) \times \prod_{i=2}^{k} P(w_i | [w_{n-i+1} ... w_{i-1}]) \right)^{-\frac{1}{k}}$$

- Looking at the above product of probabilities, we notice that if only one probability is 0, the whole product is $\frac{1}{0} = +\infty$!
- This would happen if the test corpus contains **one *n*-gram that the training corpus does not have** – which is far from unlikely!
- So perplexity is **unlikely to distinguish good from bad *n*-gram models**, as we defined them. Both kinds will likely get an infinite perplexity.
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A solution: smoothing

- To prevent perplexity from vanishing, we can simply **add a small increment** to all our n-gram counts!

$w_i \rightarrow$ $\downarrow w_{i-1}$	<s>	my	dog	is	nice	cat	nasty	your	hates	</s>	Total
<s>	ϵ	2+ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	1+ϵ	ϵ	ϵ	3+10ϵ
my	ϵ	ϵ	1+ϵ	ϵ	ϵ	2+ϵ	ϵ	ϵ	ϵ	ϵ	3+10ϵ
dog	ϵ	ϵ	ϵ	1+ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	1+10ϵ
is	ϵ	ϵ	ϵ	ϵ	1+ϵ	ϵ	1+ϵ	ϵ	ϵ	ϵ	2+10ϵ
nice	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	1+ϵ	1+10ϵ
cat	ϵ	ϵ	ϵ	1+ϵ	ϵ	ϵ	ϵ	ϵ	1+ϵ	1+ϵ	3+10ϵ
nasty	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	1+ϵ	1+10ϵ
your	ϵ	ϵ	ϵ	ϵ	ϵ	1+ϵ	ϵ	ϵ	ϵ	ϵ	1+10ϵ
hates	ϵ	1+ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	1+10ϵ
</s>	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	ϵ	-
Total	-	3+10ϵ	1+10ϵ	2+10ϵ	1+10ϵ	3+10ϵ	1+10ϵ	1+10ϵ	1+10ϵ	3+10ϵ	

- How do you think increasing/decreasing ϵ will affect training set perplexity? Test set perplexity?

How smoothing affects conditional probabilities

$$P(w_i|[w_{i-n+1}, ..., w_{i-1}]) \simeq \frac{C_{smooth}([w_{i-n+1}, ..., w_i])}{C_{smooth}([w_{i-n+1}, ..., w_{i-1}])}$$
$$\simeq \frac{C([w_{i-n+1}, ..., w_i]) + \epsilon}{C([w_{i-n+1}, ..., w_{i-1}]) + \epsilon|L|}$$

- With $|L|$ the size of the lexicon (=set of unigrams).
- Why do we get $\epsilon|L|$ in the denominator and not just ϵ?
- Recall that the number of occurrences of a $(n-1)$-gram can be expressed as the number of occurrences of n-gram continuations.

$$C_{smooth}([w_{i-n+1}, ..., w_{i-1}]) = \sum_{w \in L} C_{smooth}([w_{i-n+1}, ..., w_{i-1}, w])$$
$$= \sum_{w \in L} (C([w_{i-n+1}, ..., w_{i-1}, w]) + \epsilon)$$
$$= \sum_{w \in L} C([w_{i-n+1}, ..., w_{i-1}, w]) + \epsilon|L|$$
$$= C([w_{i-n+1}, ..., w_{i-1}]) + \epsilon|L|$$

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&\simeq \frac{C([w_{i-n+1}, ..., w_i]) + \epsilon}{C([w_{i-n+1}, ..., w_{i-1}]) + \epsilon|L|}
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&= \sum_{w \in L} C([w_{i-n+1}, ..., w_{i-1}, w]) + \epsilon|L| \\
&= C([w_{i-n+1}, ..., w_{i-1}]) + \epsilon|L|
\end{aligned}
$$

Another solution: stupid backoff

- Instead of messing with all the n-gram counts, one can revert to a $(n-1)$-gram model, just in case the n-gram probability is 0.
- If for instance $P(\text{great}|\text{I had such a})$ is 0, then maybe $P(\text{great}|\text{had such a})$ is not? And if it is 0 too, then let's try $P(\text{great}|\text{such a})$! Etc.
- This kind of hybrid model can be expressed in the following, recursive[3] way (λ being some discounting factor).

$$SB(w_i|[w_{i-n+1}, ..., w_{i-1}]) = \begin{cases} P(w_i|[w_{i-n+1}, ..., w_{i-1}]) \text{ if } > 0 \\ \lambda SB(w_i|[w_{i-n+2}, ..., w_{i-1}]) \text{ otherwise} \end{cases}$$

- This model is no longer a proper probability distribution. Additionally, it requires us to compute k-gram occurrences for potentially every $k < n$!
- But unlike the smoothing solution, it may assign different probabilities to unlikely vs. super unlikely sequences.

[3] Notice how the "otherwise" case calls SB on a smaller context.

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Lab

https://colab.research.google.com/drive/1nVcnoWj_YUDUEE_5flRhk-wnZ9WVDnNRb?usp=sharing